

Retrieval-Augmented Generation Question Answering System for Post-COVID Conditions Patient Education

Gusti Ngurah Satya Bagus Partama^{a1}, Desy Purnami Singgih Putri^{a2}, Latifa Dwiyanthi^{b3}

^aDepartment of Information Technology, Faculty of Engineering, Universitas Udayana,
Jl. Raya Kampus Udayana, Jimbaran, Bali, Indonesia,

¹ngurahsatyabagus@unud.ac.id,

²desysinggihputri@unud.ac.id (Corresponding author)

^bGraduate School of Natural Science and Technology, Division of Electrical Engineering and
Computer Science, Kanazawa University

Kanazawa, Ishikawa, Japan

³latifa@stu.kanazawa-u.ac.jp

Abstract

Post-COVID conditions is a chronic health condition emerging at least four weeks after acute COVID-19. Due to its prolonged nature, patient education and self-management are crucial for mitigating long-term symptoms, though their implementation faces challenges. To address this, our study developed an LLM-based question answering system using a retrieval-augmented generation (RAG) approach. This study compared three scenarios: naive RAG, advanced RAG, and without RAG. Evaluations on 20 verified question-answer pairs measured three aspects: answer quality, RAG retrieval quality, and systems duration. Answer quality showed no statistically significant differences across the three systems. Similarly, the RAG retrieval quality had no significant difference, despite advanced RAG showing a significantly better Euclidean distance score for its search results. Regarding duration, advanced RAG was the slowest compared to the other systems. In the end, the naive RAG adaptation was concluded to be the best developed system.

Keywords: Large Language Model, Patient Education, Post-COVID Conditions, PCCs, Retrieval-Augmented Generation

1 Introduction

Post-COVID conditions (PCCs) are a health condition that begins at least four weeks after the initial infection with COVID-19. Also known as post-acute sequelae of SARS-CoV-2 infection (PASC) or long COVID-19, it can manifest with a variety of neurological and psychiatric symptoms [1], [2]. This condition is classified as chronic, with a significant survivability burden comparable to cancer in terms of physical, psychological, social, and economic impacts on infected individuals and their families [3]. According to a report, 25% of PCC-positive individuals showed evidence of damage to multiple organs, including the heart, lungs, kidneys, and pancreas, after previously receiving hospital care for COVID-19 [4]. Furthermore, 66% of low-risk and younger patients also experienced damage to one or more organ systems. Therefore, it can be considered as serious as an acute COVID-19 infection.

Given the chronic nature of PCCs, patient education and strengthening self-management skills

are crucial aspects of efforts to mitigate long-term symptoms [5]. Patient education is the provision of information and knowledge services based on the patient's health condition, including education about presurgical procedures, to empowerment-focused education aimed at improving recovery results and treatment results [6]. Despite its importance, patient education still faces many challenges due to factors such as a lack of coordination between healthcare departments, insufficient training for healthcare and nursing staff, and a lack of commitment from healthcare administration [6]. In the case of PCCs patient education, according to Garg et al. [7], the main challenges lie in the complexity of health conditions, the high burden of symptoms in patients with PCCs and the need for continuous intensive care. In addition, it is important for patient education to include interactive education focused on empowering patients [6], [7].

The implementation of state-of-the-art technology is necessary to achieve patient education and strengthen the self-management capabilities of PCCs patients. ChatGPT as a health education tool shows potential in providing general knowledge, but it is still poor at providing medical recommendations compared to Google search engines [8], indicating the need for a more sophisticated approach to support patient education. In response to this challenge, Wang et al. [9] developed an optimization-based approach to LLMs (large language models) using a framework called RISE (Retrieval-augmented Information System for Enhancement), which is an extension of RAG method designed to optimize LLMs' ability to answer more specific medical questions, particularly for diabetes patient education. The implementation of the framework demonstrated improvements in accuracy and comprehensiveness for LLM models (GPT-4, Claude 2, Google Bard), with an average increase in response accuracy of 12% (15/129) compared to the performance of LLMs without the RISE system. Specifically, for each developed model system, GPT-4 saw an increase of 7% (3/43), Claude-2 saw an increase of 19% (8/43), and Google Bard saw an increase of 9% (4/43). The RISE framework also improved the comprehensiveness and understandability of LLMs with an average increase of 0.44 (SD 0.10) and 0.19 (SD 0.13), respectively, as evaluated by three medical experts using a scoring system from one to five. These results demonstrate the significant potential of implementing RAG-based optimization technologies, such as RISE, to improve the quality of patient education systems, making them more accurate, comprehensive, and easy to understand.

This study develops an interactive patient education solution for PCCs, focusing on strengthening patients' self-management skills. The solution involves the development of a closed-domain question-answering system by comparing three methods: without RAG, naive RAG, and advanced RAG. The primary objective of this study is to explore the best RAG method to enable LLMs to provide more reliable, accurate, and traceable medical answers. Furthermore, this study also addresses methodological gaps in the development of LLM-based QA systems for the medical domain. The research offers a comprehensive experimental approach that compares various RAG architectures, which were limited in previous studies, to generate accurate, contextual, and efficient answers.

2 Research Methods

The research adopted a four-stage workflow. These stages included document acquisition, model selection, system development, and system evaluation as shown on Figure 1. The process of document acquisition began by searching for medical journal articles on PCCs using the keyword of "post-COVID-19 syndrome or post-acute sequelae of SARS-CoV-2 infection and PACS" from PubMed literature database. An initial search on 12 May 2025 identified a total of 150 potentially relevant publications. This number was obtained after filtering the results to include language filters to English only, limited to only free and open access articles, certified MEDLINE, and having specific article types, such as case reports, clinical studies, clinical trials, meta-analyses, guidelines, reviews, systematic reviews, and patient education handouts. Following the deduplication process, the initial collection was reduced to 100 unique and relevant publications, which were then prepared for data extraction using Selenium to simulate the process of PDF acquisition. Acquired PDF documents were directly funneled from the download source to Google Cloud Storage, leveraging Cloud Storage principles for efficient data ingestion.

With document acquisition complete, the study moved to the model selection phase. The primary objective was to identify the most suitable language models for two key components, the question-answering system's answer generator and the advanced architecture's pre-retrieval (query rewriter) and post-retrieval (context summarization) components. For answer generation, the DeepSeek R1 model (8 billion parameters) was chosen for its robust reasoning capabilities, which are well-suited for complex medical applications. This model was accessed via the open-source Hugging-Face inference platform. In contrast, the more demanding tasks of query rewriting and context summarization utilized the state-of-the-art OpenAI GPT-4o model. Additionally, an embedding model was selected in this phase. The text-embedding-ada-002 model was chosen for the text-to-vector transformations needed for the RAG system's retrieval components. Accessed through the OpenAI API, this model converted both document chunks and user queries into 1536-dimensional vectors [10].

Following the model selection phase, the third stage of the workflow, system development, was initiated. This phase involved the construction of three distinct question-answering systems for comparative analysis, an advanced RAG, a naive RAG, and a no-RAG baseline. The development of these components was performed using the Python LangChain library. This library facilitated the application of prompt engineering and few-shot learning techniques to the answer generation, pre-retrieval, and post-retrieval components.

A comprehensive evaluation marked the final stage of the workflow, designed to assess the performance of the three distinct question-answering systems. The final evaluation was conducted using a dedicated instrument comprising 20 pairs of questions and corresponding references. This instrument was designed to measure three key performance indicators, namely answer quality, retrieval quality, and system process duration.

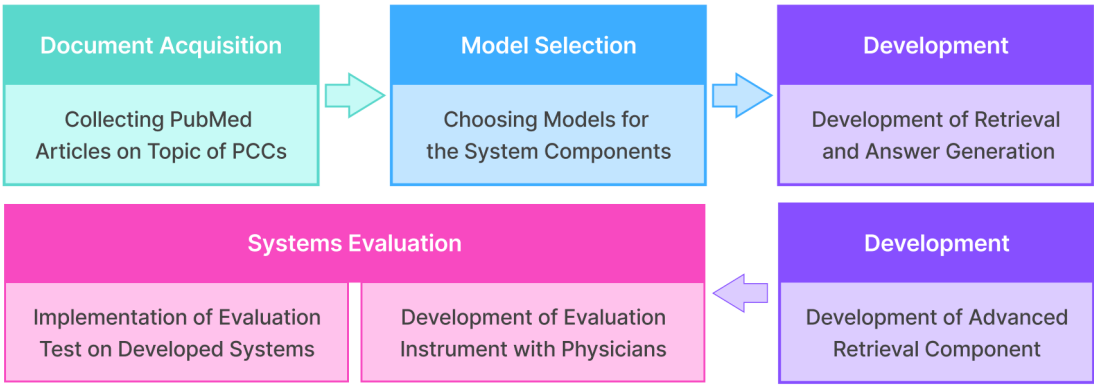


Figure 1. Research Process

The evaluation instrument was developed in collaboration with a pulmonologist specialist. This instrument, consisting of 20 question-reference pairs, was inspired by an evaluation instrument from a previous study on RAG adaptation on question answering system for diabetes patient education [9]. The questions were organized into five distinct categories, with four questions in each category. These categories included concept of PCCs, symptom and causes of PCCs, test and diagnosis of PCCs, PCCs management, and PCCs prevention. The pulmonologist assisted in selecting the appropriate questions from each of the categories and subsequently confirmed and verified the corresponding answers. This verification process involved making additional adjustments to some questions to ensure clinical precision and relevance. The instrument was ultimately designed to ensure the clinical relevance of the questions and their corresponding answers, providing a robust test set for the system's performance.

The system architecture for this study was adapted from the framework proposed by Gao et al [11], which employs a chain of distinct components for the naive RAG and advanced RAG architecture. A visual representation of the difference of the complete system is provided in Figure 2.

The end-to-end design of the three system architectures is illustrated in the accompanying figure

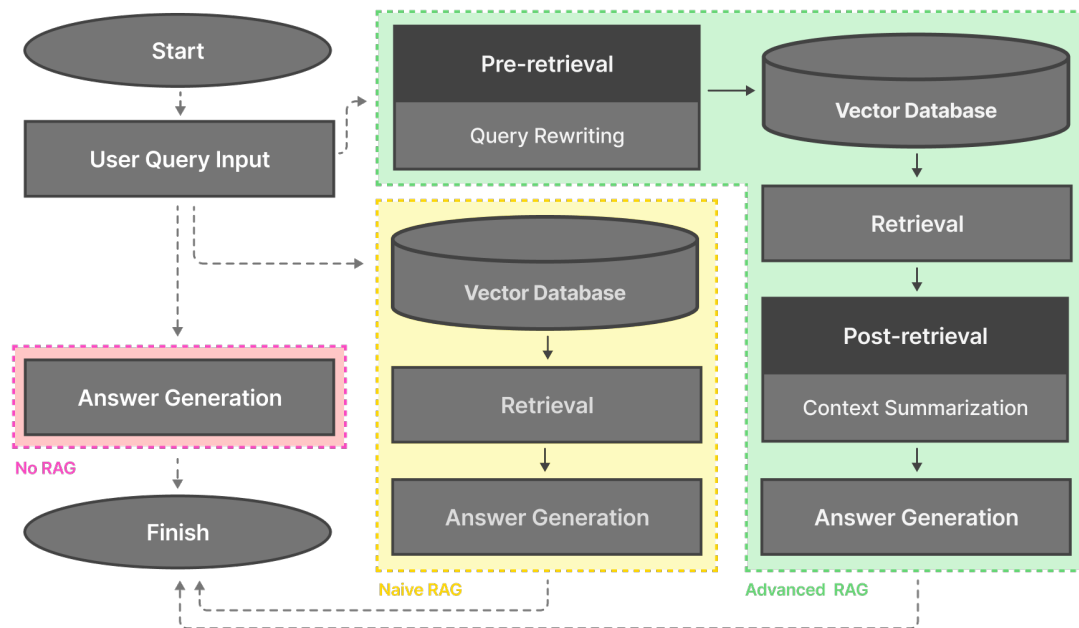


Figure 2. Question Answering Systems Design

2. The No-RAG system is depicted in red, the Naive RAG in yellow, and the Advanced RAG in green. The No-RAG architecture consisted solely of an answer generation component which developed with different prompt engineering from the RAG systems. In contrast, the Naive RAG system incorporated both a retrieval component from the vector database and an answer generation module. The Advanced RAG system featured a more complex design, integrating additional pre-retrieval and post-retrieval components prior to the final answer generation.

2.1 Pre-retrieval Component

The pre-retrieval component, which is executed before the retrieval process, is an additional element of the advanced RAG system architecture. This component focuses on transforming patient queries into a more complex and technical format, correcting typographical errors, and making informed assumptions about patient questions to facilitate a more educational response. Its primary goal is to improve the retrieval score and thereby obtain more relevant search results compared to the Naive RAG system. This adaptation is implemented using a few-shot learning approach with the OpenAI GPT-4o language model.

2.2 Retrieval Component

The knowledge source used by the RAG system's retrieval component is a local FAISS vector database, which was constructed from 100 successfully acquired PDF documents. The documents were prepared for indexing using a recursive chunking approach [12], which segments the information into smaller, more manageable parts to facilitate the retrieval of document sections directly related to the user's query. The goal is to obtain the top three relevant documents ($k=3$) based on a retrieval score measured using the Euclidean distance of the FAISS calculations. This same principle is applied to both the Naive RAG and Advanced RAG systems. A total of 13,807 chunks were generated, each with a length of 512 characters and an overlap of 51 characters, with the overlap implemented to ensure that the information breakdown did not create gaps in

important content.

Retrieval was implemented using the LangChain library's `similarity_search_with_score` module. This process uses the Euclidean distance for similarity calculations, which is the default method in FAISS. Vectorization for this process was performed using the `text-embedding-ada-002` model, accessed via the Open AI API. Although both the naive RAG and advanced RAG systems use this retrieval process to obtain the top three most relevant document chunks, they differ in their query input. The naive RAG system utilizes the user's original query, whereas the advanced RAG system uses the rewritten query generated by the pre-retrieval component. The Euclidean distance formula used by OpenAI for this process is described as follows.

$$d^2(Q, P) = \sum_{i=1}^n (q_i - p_i)^2 \quad (1)$$

Equation 1 illustrates the similarity calculation performed by the FAISS Euclidean module. The squared Euclidean distance between the query vector Q and a passage vector P is represented by the variable $d^2(Q, P)$. This calculation is obtained by summing the squared differences of each corresponding vector dimension. This adaptation uses vectors with a length of 1.536 dimensions, which corresponds to the output length of the `text-embedding-ada-002`.

2.3 Post-Retrieval Component

The post-retrieval component is an additional element of the advanced RAG system, executed after the retrieval process is complete. The primary function of this component is to summarize the retrieved context into a clearer, more concise, and noise-free format. Similar to the pre-retrieval component, it is adapted using a few-shot learning approach with the GPT-4o model and implemented via the Python LangChain library. This process is implemented to reduce the workload on the advanced RAG answer generation system and minimize input tokens, thereby preventing model hallucinations caused by lengthy input context.

2.4 Answer Generation

The system components were developed using distinct prompts for the RAG and non-RAG architectures. A few-shot learning approach was applied to both, with the key difference being that the RAG systems incorporated a chain of retrieved results, whereas the non-RAG system did not. The DeepSeek R1:8b Qwen reasoning model was selected as the answer generation component for all three systems. This model was chosen to provide more comprehensive answers through customized prompt settings tailored for PCCs patient education. The Novita Inference API on the HuggingFace platform was implemented to enable the use of this model across all three systems, ensuring consistent answer generation capabilities.

2.5 Evaluation

The system evaluation was conducted as a means of measuring the performance of the three question answering systems that had been developed. The measurements in this study focused on the categories of system answer quality, retrieval quality specific to the RAG system, and processing time efficiency during testing. The METEOR (Metric for Evaluation of Translation with Explicit ORdering) and RAGAS (Retrieval-augmented Generation Assessment) Faithfulness measurement approaches were selected to evaluate the performance of system answer quality. Additionally, RAGAS was used in the Faithfulness and Context Recall metrics for evaluating the retrieval performance of the RAG system, along with a comparison of Euclidean retrieval scores. Finally, for calculating system efficiency, a comparison in seconds was applied to the processing time from when the answer was entered until all processing was completed. The complete formula and its adaptation are explained as follows.

1. METEOR (Metric for Evaluation of Translation with Explicit Ordering) is a statistical evaluation metric for machine translation developed as a method to address the shortcomings of the BLEU evaluation method developed by IBM [13]. The calculation involves several steps:

- (a) Word Alignment of the system generated and reference based on exact matches, stems, synonyms, and paraphrases, ensuring each word in the system generated and reference sentences is used only once to maximize the overall matched.
- (b) Precision and recall calculation. Precision used to measure the proportion of matched words in system generated compared to the total number of words in system generated, while recall used to measure the proportion of matched words in system generated compared to the total number of words in reference.

$$Precision (P) = \frac{Number\ of\ words\ matched}{Number\ of\ words\ in\ response} \quad (2)$$

$$Recall (R) = \frac{Number\ of\ words\ matched}{Number\ of\ words\ in\ reference} \quad (3)$$

- (c) Penalty calculation by multiplying the proportion of number of chunks matched between the system generated to reference and the number of total matched words.

$$Penalty = 0.5 \times \frac{Number\ of\ chunks}{Number\ of\ words\ matched} \quad (4)$$

- (d) METEOR score score is calculated using the harmonic mean of precision and recall, adjusted by the penalty factor.

$$METEOR\ Score = F_{mean} \times (1 - Penalty) \quad (5)$$

where

$$F_{mean} = \frac{10PR}{R + 9P} \quad (6)$$

The calculation of METEOR score is implemented using the NLTK library's with meteor_score function with help of Wordnet to help the word alignment process in finding words synonyms. METEOR having range of score from 0 to 1, with the higher the score indicating a better semantic lexical alignment between the system generated answer and reference.

2. RAGAS (Retrieval-augmented Generation Assessment) is an evaluation metric that works semantically and contextually on the relationship between RAG system components using the LLM-as-a-judge approach in the process [14]. The RAGAS method adapted in this study consists of three types, namely, faithfulness, answer correctness, and context recall.

- (a) Answer Correctness used to evaluate the system answer quality by calculating the proportion of numbers of true positive statements and calculation of true positive statements with additional of false positive and false negative statements multiplied by 0.5 as shown below.

$$Answer\ Correctness = \frac{|TP|}{(|TP| + 0.5(|FP| + |FN|))} \quad (7)$$

- (b) Faithfulness used to evaluate the system retrieval quality by calculating the system proportions of claims number in the system generated supported by the retrieved context and the total number of statement in the system generated. Reflecting the alignment of statement number in retrieved context and system generated answer.

$$Faithfulness = \frac{Number\ of\ claims\ in\ the\ response\ supported\ by\ the\ retrieved\ context}{Total\ number\ of\ claims\ in\ the\ response} \quad (8)$$

- (c) Context Recall used to evaluate the system retrieval quality by calculating the system proportions of claims number in the reference supported by the retrieved context and the total number of statement in reference. Reflecting the alignment of statement number in retrieved context and reference.

$$\text{Context Recall} = \frac{\text{Number of claims in the reference supported by the retrieved context}}{\text{Total number of claims in the reference}} \quad (9)$$

3 Result and Discussion

3.1 Result

The evaluation results are presented in swarm boxplots, with categories for answer quality, retrieval quality, and system processing time. These results were obtained from a single experiment where each system was tested on a set of 20 questions. The questions were organized into five categories, each containing four questions.

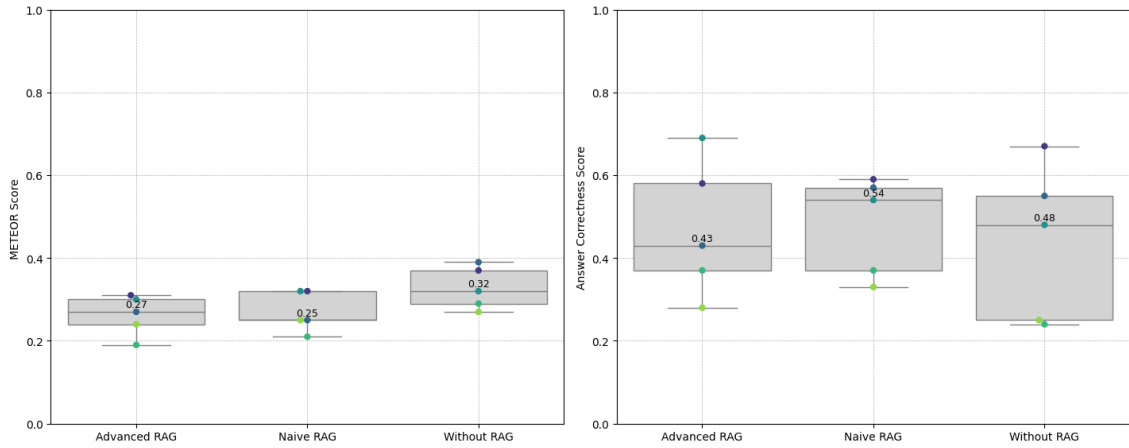


Figure 3. Answer Quality

Figure 3 compares the evaluation results using the METEOR and RAGAS answer correctness metrics for assessing answer quality. The METEOR evaluation shows that the system without RAG has the highest median value at 0.32, outperforming both the naive RAG and advanced RAG systems. However, while measuring a similar parameter, the RAGAS answer correctness evaluation shows that the naive RAG system has the highest median value at 0.54. This indicates that the METEOR metric, which operates on a lexical-semantic basis, finds the system without RAG to have the best score, whereas the RAGAS answer correctness metric, which uses a contextual approach, favors the naive RAG system.

Figure 4 visualizes the evaluation of RAG system retrieval quality, comparing results across the Euclidean retrieval score, RAGAS faithfulness, and RAGAS context recall metrics. The evaluation shows that the naive RAG system had a higher median score for RAGAS faithfulness (0.38) and RAGAS context recall (0.14) compared to the advanced RAG system. This contrasts to the median retrieval score, where the advanced RAG system performed better with a value of 0.23. This indicates a conflict between the Euclidean retrieval score and the RAGAS metrics. The results suggest that while pre-retrieval techniques can improve retrieval scores, they do not necessarily guarantee that the final answer will be aligned with the retrieved search results or references.

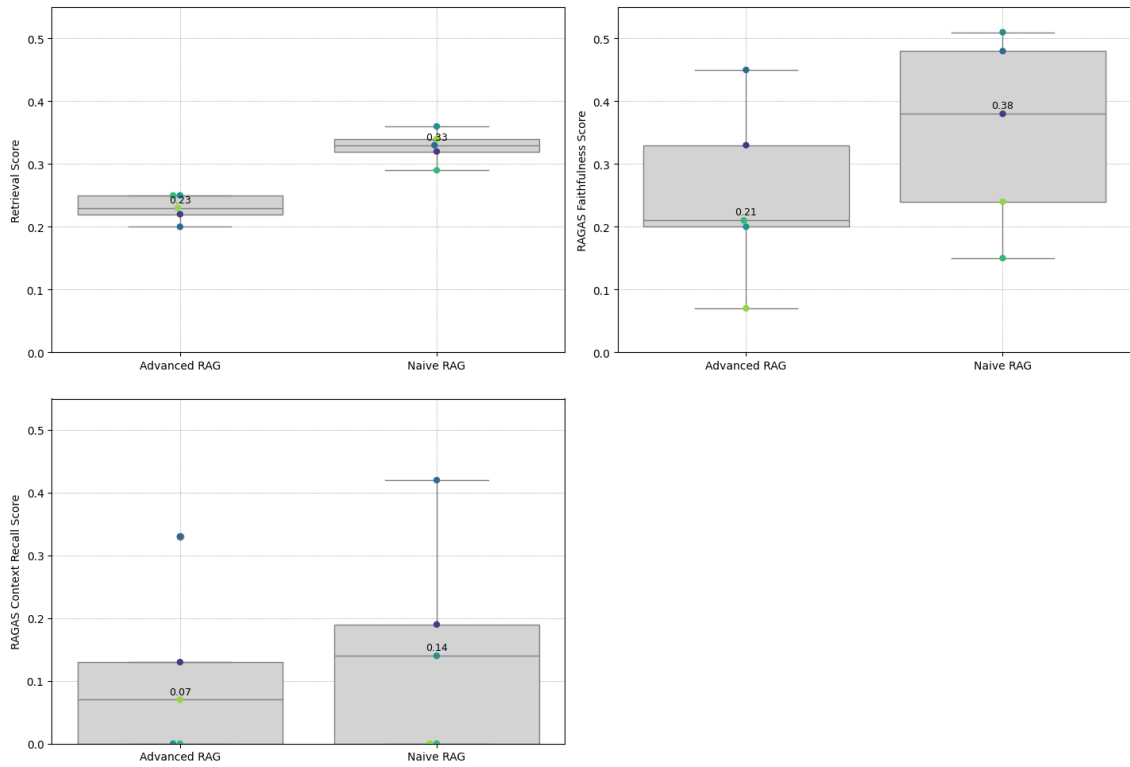


Figure 4. Retrieval Quality

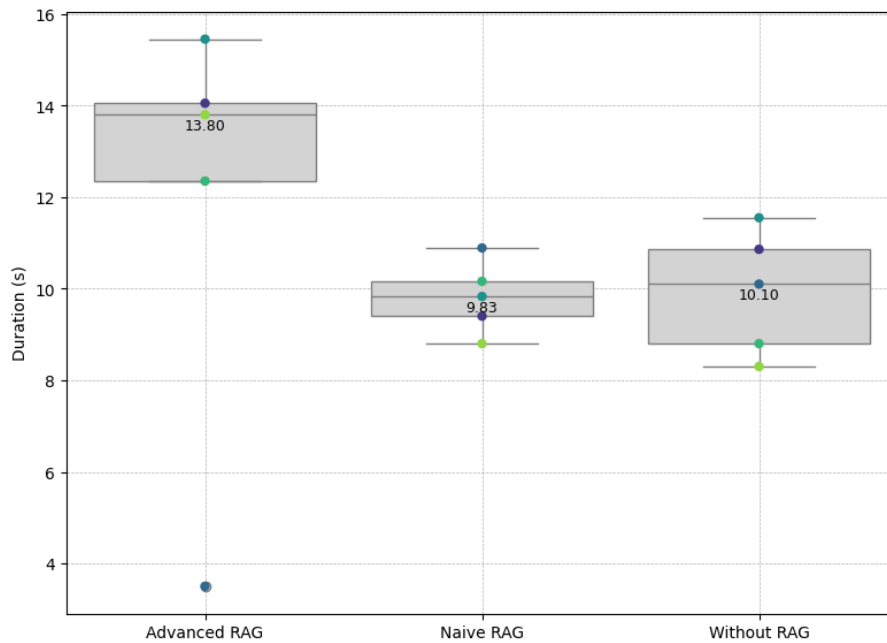


Figure 5. Systems Processing Duration

Figure 5 presents a comparison of the end-to-end processing duration for each system. As expected, the advanced RAG system recorded the longest processing time, with a median of 13.80 seconds, which is attributable to its more complex processing chain. However, a notable finding is that the system without RAG was the fastest, with a median of 9.83 seconds, outperforming the

naive RAG system, which had a median of 10.10 seconds. This indicates that the addition of the retrieval process in the naive RAG system significantly impacts processing time, making it slower than the base generative model.

Metrics	Test Type	Stats.	Sig.	Conclusion
METEOR	Friedman Chi-square	F = 0,5600	0,0254	Non Significance
Answer Correctness	Friedman Chi-square	F = 7,3421	0,7558	Non Significance
Faithfulness	Wilcoxon	T = 84,5000	0,4440	Non Significance
Context Recall	Wilcoxon	T = 12,0000	0,7353	Non Significance
Retrieval Score	Friedman Chi-square	T = 0,0000	0,0001	Significance
Duration (s)	Friedman Chi-square	F = 19,6000	0,0001	Significance

Table 1. Significance Test Result

Table 1 summarizes the statistical significance analysis for all evaluation metrics. Using a Wilcoxon Signed-Rank test for pairwise comparisons between the naive RAG and advanced RAG systems, and a Friedman Chi-square test for all three systems, the analysis revealed a mix of findings. For answer quality, there was no statistically significant difference across the systems on either the METEOR or RAGAS answer correctness metrics, indicating they all produced answers of a comparable quality. In terms of retrieval, while a significant difference was found for the Euclidean retrieval score which favored the advanced RAG system, no significant difference was observed between the naive and advanced RAG systems for RAGAS faithfulness or RAGAS context recall. This suggests that while pre-retrieval may improve a system’s ability to find relevant documents, it doesn’t necessarily improve the factual accuracy or contextual alignment of the final answer. The most definitive finding was in system process duration, where a significant difference was clearly identified: the advanced RAG system was the slowest with a median of 13.80 seconds, while the system without RAG was the fastest at 9.83 seconds.

3.2 Discussion

Effective question-answering systems are evaluated based on their answer quality, retrieval accuracy, and computational efficiency through end-to-end process duration. To assess the quality of the responses, we utilized two distinct metrics. METEOR provides a statistical semantic-lexical perspective, while RAGAS answer correctness focuses on semantic-contextual alignment. The systems achieved a median METEOR score of 0.32 and a RAGAS answer correctness score of 0.54, respectively, for the no RAG and naive RAG implementations. Despite these score differences, the results of a significance test showed no statistically significant difference between the systems, suggesting that, based on the metric of answer quality, there is no clear superior system. However, a comparison with the findings of Firdaus et al [15] shows that our METEOR scores are notably higher, with our no RAG system’s score of 0.32 exceeding their best average score of 0.22 by approximately 45%. This elevated performance, although not statistically significant within our benchmark, indicates a strong result relative to existing research on similar technologies. This discrepancy can be attributed to two key factors. One is the size of the dataset, as this study used 100 documents compared to 9 documents in the other research. Another is the choice of LLM, with this study employing DeepSeek R1;8b and the other research using Llama 7b as the answer generation component.

The analysis of system retrieval quality, using the RAGAS context recall and RAGAS faithfulness metrics, also yielded non-significant results in terms of overall system performance. However, a significant difference was observed in the euclidean retrieval scores for the advanced RAG system, indicating a more aligned context of the queries and the retrieved context chunks. This suggests that the pre-retrieval component of the advanced RAG, which polishes a simple question into a more complex and detailed query, is effective. Despite the success of its pre-retrieval component, the advanced system’s overall answer quality did not improve. This is likely due to the poor performance of the post-retrieval stage. This suggests that the current context summa-

rization method is not effectively enhancing the final answer, and alternative approaches, such as reranking, may be necessary to be explored in further research.

System efficiency constitutes a third key dimension for evaluating these systems, where a balance between performance and quality is paramount. In terms of process duration, the naive RAG system was the fastest, at 9.83 seconds. This was followed by the no RAG system at 10.10 seconds, and the advanced RAG system at 13.80 seconds. Subsequent significance testing confirmed that the advanced RAG system's duration was significantly slower than the other two. In contrast, the difference in processing time between the no RAG and naive RAG systems was not statistically significant, placing them both in the category of the fastest systems.

4 Conclusion

Based on a comprehensive analysis of the developed question-answering systems, a number of key trade-offs and conclusions can be drawn. The evaluation of answer quality revealed a notable paradox. While different metrics showed conflicting top performers, a statistical analysis found no significant difference across all three systems. This suggests that for a specialized domain like patient education, the added complexity of a RAG pipeline may not yield a significant gain in overall answer quality.

A critical trade-off was identified in retrieval quality. Although the advanced RAG system showed a significantly better Euclidean distance score for its search results, indicating improved document retrieval accuracy, this did not lead to a significant difference in RAGAS faithfulness or context recall. This highlights a key limitation, as a technical improvement in retrieval does not necessarily translate into a more factually correct or contextually aligned final answer.

Finally, a clear trade-off between performance and efficiency was confirmed in system duration. The advanced RAG system had the longest processing time, while the naive RAG system was the fastest, a result confirmed as significantly quicker than the others. Considering these findings, the naive RAG adaptation represents the optimal solution, as it successfully provides the core benefits of a RAG pipeline without the significant computational overhead of the advanced system, striking the best balance of performance and efficiency.

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